

Perception Pressure and Resonance

Decomposable, Trust-Weighted Measures of Media Narrative from a Multi-Perspective News-Aggregation System

prism_preprint.tex · 7pp · Jun 2026 · system lives at ~/Prism (separate repo) · lead preprint for Leippold (UZH) · methods/system paper, no empirics

Thesis

Two quantitative measures of media narrative computed continuously from a live multi-source news stream, plus the system that produces them. Not another sentiment index: the design goal is a measure that is *signed*, *magnitude-meaningful*, *manipulation-resistant*, and *fully auditable*.

Resonance — per-story media impact

$$\text{Res}(c, t) = \underbrace{\text{breadth}(c)}_{\text{diversity multiplier}} \cdot \sum_{m \in c} \text{trust}(s_m) \text{eng}(m) \text{dec}(a_m)$$

with $\text{breadth}(c) = \log_2(1 + n_{\text{sources}})$, $\text{eng}(m) = 1 + \log_2(1 + \text{re}_m/\text{median})$, $\text{dec}(a) = e^{-\lambda a}$, $\lambda = \ln 2/h$.

Perception Pressure — per-concept longitudinal signal

salience: $A(K, t) = \sum_c \text{breadth}(c) \text{dec}(a_c) \geq 0$

valence: $V(K, t) \in [-1, +1]$ (credibility-weighted)

perception: $P(K, t) = A(K, t) \cdot V(K, t)$

momentum $\dot{P}(K, t) = P(K, t) - P(K, t - 1)$.

Three properties, novel in combination

1. **Multiplicative coupling** of attention and directional sentiment $\Rightarrow$ the signal is *signed* and its magnitude means something. Volume-only proxies have no sign; sentiment-only indices have no weight.
2. **Diversity as a multiplier, not a sum** $\Rightarrow$ a single outlet cannot inflate the measure. Multi-perspective by construction, not by editorial policy.
3. **Every component persisted independently** $\Rightarrow$ fully decomposable and auditable. You can always answer “why did P move” — it is not a black box.

Trust is earned, not assigned

$$\text{trust} = 0.1 + 0.4 \cdot \frac{\# \text{ validated}}{\# \text{ total}}$$

via an automated cross-corroboration lifecycle. **No hand-assigned credibility table** — which is where comparable indices smuggle in the researcher’s politics, and the first thing a referee will probe.

The system

Five-agent pipeline — *discover* $\rightarrow$ *analyze* $\rightarrow$ *track* $\rightarrow$ *personalize* $\rightarrow$ *deliver* — coordinated by a database state machine. Prism is the upstream producer of the narrative/hype series NAT ingest.

Positioning

Against volume-only attention proxies (Google Trends, article counts) and direction-only sentiment indices. The claim is not

“better sentiment” but a *different object*: a signed pressure with interpretable units and an audit trail.

Downstream application (specified, not claimed)

Treat $P(K, t)$ as an **exogenous narrative signal** whose incremental predictive value for financial returns is measured under a leakage-controlled information-theoretic protocol. [!] The paper is explicit that this is a methods-and-system paper: the instrument is fully specified and parameterized for ablation; **empirical validation is the next step, not a claim made here**. Neither Prism metric has logged results. Do not let a reader infer otherwise.

Why it is honest to publish results-free

Same posture as #3 and #5: specify the instrument, expose the constraint, then measure. A referee can attack the construction on its merits without having to disentangle it from a performance claim — and there is nothing to retract when the empirics land.

The five agents

Coordinated not by a message queue but by a **database state machine**: RAW $\rightarrow$ ANALYZED $\rightarrow$ DELIVERED.

Discovery	pull + cluster articles by event; grow the source registry (RSS ingest; lexical dedup — Jaccard, TF-IDF, entity overlap)
Analysis	summarize a cluster; extract per-source perspective, sentiment, bias, claims (LLM structured extraction + quality score)
Tracking	scan clusters for tracked concepts; compute the measures
Personalize	per-reader selection
Deliver	emit

Stated editorial principle: make bias *visible* rather than hidden — no original reporting, every claim attributed, multiple framings side by side. The quantitative measures are instrumentation of that principle, not a bolt-on.

Disclosed threats to validity

[!] Sentiment/bias labels come from an **LLM, not gold-standard human annotation** — the structure of that measurement error is itself an open research question, and it is the first thing a referee will press. Clustering is **lexical, not embedding-based** (token overlap, TF-IDF, regex entity extraction), so events can split or merge. The engagement term **defaults to neutral** when reaction counts are missing, so it is often inert in practice. Storage is single-writer, bounding throughput. None of these undermine the *definitions*; all of them bear on the fidelity of the computed series and must be controlled in any empirical study.

Siblings

#5 (*From Narrative to Alpha*) is the direct sequel: it takes $P(K, t)$ as given and argues, at the level of methodology, what an exogenous signal contributes to a microstructure book — timescale separation, effective breadth, partial evasion of the adverse-selection barrier. Send #4 then #5 to Leippold; #4 alone reads as a systems paper.